(3) Apply Huffman Algorithm to build to huffman Tree:-

B[1] - G[1] - S[1] - C[2] - O[2] - R[2] - Y[2] - E[3] - I[3] - L[4] - N[4] - A[7]

(1) combine B[1], G[1] → BG[2]

S[1] - C[2] - D[2] - O[2] - R[2] - Y[2] - BG[2] - E[3] - I[3] - L[4] - N[4] - T[4] - A[7]

(2) combines[1] and C[2] → SC[3]

D[2] - O[2] - R[2] - Y[2] - BG[2] - E[3] - I[3] - SC[3] - L[4] - N[4] - T[4] - A[7].

(3) combine D[2] and O[2] to get DO[4]

R[2] - Y[2] - BG[2] - E[3] - I[3] - SC[3] - L[4] - N[4] - T[4] - DO[4] - A[7]

(4) combine R[2] and Y[2] to get RY[4]

BG[2] - E[3] - I[3] - SC[3] - L[4] - N[4] - T[4] - DO[4] - RY[4] - A[7]

(5) BG[2], E[3] → BGE[5]

I[3] - SC[3] - L[4] - N[4] - T[4] DO[4] RY[4] - BGE[5] - A[7].

(6) I[3], SC[3] to get ISC[6]

L[4] - N[4] - T[4] - DO[4] - RY[4] - BGE[5] - ISC[6] - A[7]

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(7) L[4], N[4] → LN[8]

T[4] – DO[4] – RY[4] – BGE[5] – ISC[6] – LN[8].

(8) T[4], DO[4] → TDO[8]

RY[4] – BGE[5] – ISC[6] – A[7] – LN[8] – TDO[8]

(9) RY[4] – BGE[5] → RYBGE[9]

ISC[6] – A[7] – LN[8] – TDO[8] – RYBGE[9]

10) ISC[6], A[7] → ISCA[13]

LN[8] – TDO[8] – RYBGE[9] – ISCA[13]

11) LN[8], TDO[8] → LNTDO[16]

RYBGE[9] – ISCA[13] – LNTDO[16]

12) RYBGE[9] and ISCA[13] to get RYBGETSCA[22]

LNTDO[16] – RYBGETSCA[22].

combine LNTDO[16] and RYBGEISCA[22]

to get LNTDORYBGEISCA[38]

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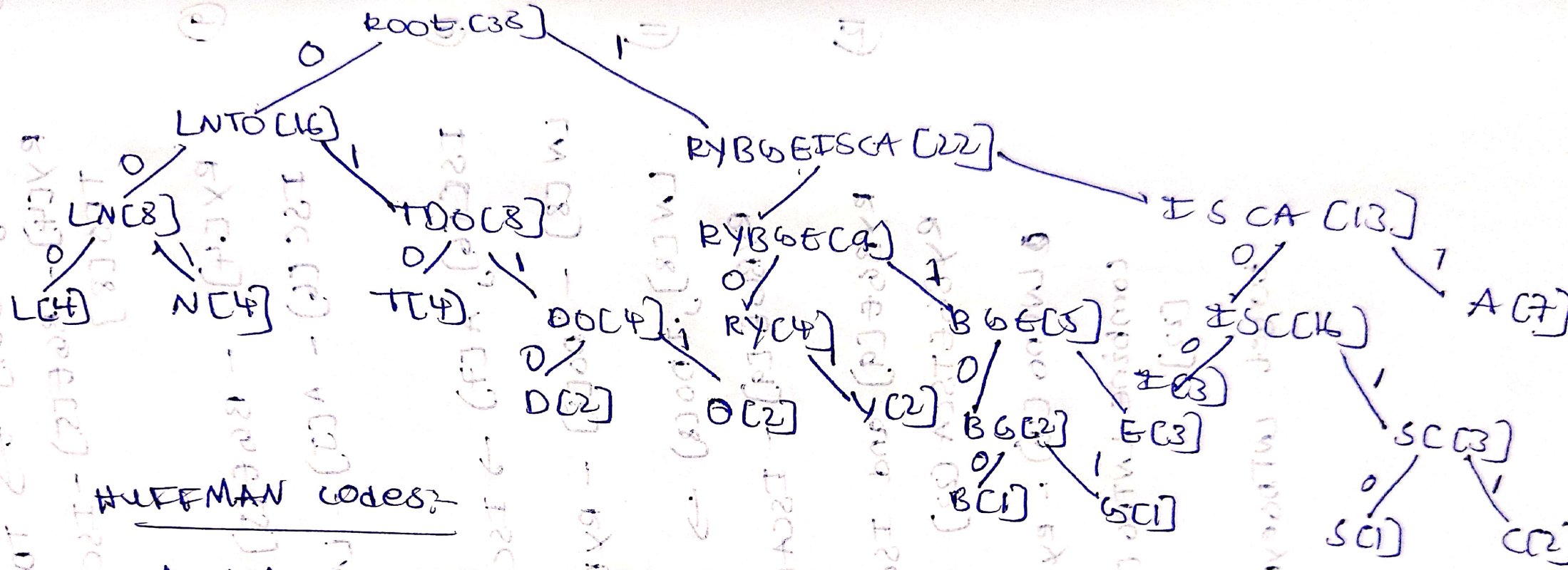

HUFFMAN codes:

A=111, L=000, N=001, T=010, G=1011, I=1100, C=1101, D=0110
O=0111, R=1000, Y=1001, B=10100, G=10101, S=11010